**Distinct analysis routes, matched to the measurement**

| | | |
|---|---|---|
| **Amplicon reads** | Control-derived background → candidate tests → empirical threshold | Retained variants; per-animal summaries |
| **Sanger clones** | Quality-trimmed bases → Q ≥ 20 changes counted | Per-clone changes; control background alongside |
| **PacBio switch reads** | Junction alignment → unique junctions → structural summaries | Microhomology and insertion phenotypes |

Control subtraction is not applied to Sanger clone counts. Switch-junction structure is a separate endpoint from substitution calling.